Darius A. Pétermann

Machine Learning Scientist specializing in generative audio, neural audio codecs, speech foundation models, and multimodal AI

Scholar Website Email LinkedIn GitHub

1 Education

Ph.D. in Intelligent Systems Engineering-Computer Science Dec 2025
Luddy School of Informatics, Computing, and Engineering, Indiana University *Bloomington, IN*

- **Selected Coursework:** *Machine Learning for Signal Processing, Deep Learning, Computer Vision, Applied Machine Learning*
- Research Group: Signals and AI Group in Engineering (SAIGE)
- Thesis: ***"Efficient Native Neural Sub-band Coding through Residual Feature Representation within Hyper-Autoencoded Reconstruction Propagation Networks"***
- Advisor: Prof. Minje Kim

M.Sc. in Sound and Music Computing Sep 2020
Dept. of Information and Communication Technologies, Universitat Pompeu-Fabra *Barcelona, Spain*

- **Selected Coursework:** *Music Information Retrieval, System Design, Audio Signal Processing, ML for Audio, Research Methods, Reinforcement Learning*
- Thesis: ***"SATB Voice Segregation for Monaural Recordings"***
- Advisor: Dr. Pritish Chandna

B.M. in Music Synthesis - Audio Engineering May 2016
Electronic Production & Design Dept, Berklee College of Music *Boston, MA*

- **Selected Coursework:** *Digital Signal Processing, Physical Computing, Audio Programming in C, Principles of Audio Electronics, Music Acoustics*
- Thesis: ***"A Deep Look at Spectral Synthesis Techniques Through csConvolve"***
- Advisor: Dr. Richard Boulanger

2 Positions Held

Netflix Dec 2025 – Present
Research Scientist *Los Gatos, CA*

- Developing expressive text-to-speech and speech-to-speech systems.
- Researching controllable generation, speaker adaptation, and neural audio codecs.
- Optimizing and evaluating speech models for low-latency production inference.

Amazon, AGI March 2025 – Nov 2025
Applied Scientist II, L5 *Bellevue, WA*

- Conducting research in the Audio Foundation Team, advancing multimodal foundation models for audio, speech, and music understanding / reasoning.
- Worked across LLM training lifecycle: large-scale pre-training and supervised fine-tuning (SFT), including evaluation and benchmarking.
- Built large-scale preprocessing pipelines (Spark) for multimodal audio–text datasets (10M+ samples) to support foundation model training.

Netflix Jun 2024 – Nov 2024
Machine Learning Research Intern *Los Gatos, CA*

- Supervised by Dr. Mahdi Kalayeh
- Working on multimodal (audio-visual) generative AI problems

Google Research May 2023 – Oct 2023
Student Researcher *Cambridge, MA*

- Supervised by Dr. Hakan Erdogan , Dr. John Herhsey , and Dr. Scott Wisdom
- Working on next-level unsupervised audio source separation problems

Mitsubishi Electric Research Labs (MERL) Summers in 2021 - 2022
Research Intern *Cambridge, MA*

- Received $15'000 of gift-money from MERL to work on the *"Cocktail Fork Problem"*
- Hosted by Dr. Gordon Wichern and Dr. Jonathan Le Roux
- Derived and implemented new models and optimization methods for audio analysis with applications to source separation in challenging multi-source environments and using advanced machine learning techniques

Signals and AI Group in Engineering (SAIGE), Indiana University Jan 2021 - Dec 2025
Research Assistant *Bloomington, IN*

- Supervised by Prof. Minje Kim
- Conducting research pertaining to neural audio coding and audio source separation problems

Apple Inc. Jun 2016 - Jul 2019
Content Engineer *Cupertino, CA*

- Software engineer for Apple's pro Audio & Music Apps (LogicPro, GarageBand)
- Designed real-time MIDI processing systems in C++ for Apple's virtual musical instruments

Electronic Production & Design Dept, Berklee College of Music Sep 2015 - May 2016
Programming Tutor *Boston, MA*

- Tutored and mentored EPD students for technical classes: *"Audio Programming in C", "Digital Signal Processing", "Csound", "Max/MSP"*

3 Publications

Preprints and Submissions

[R01] *Seeing Sound: Assembling Sounds from Visuals for Audio-to-Image Generation*, arXiv, 2025.

International Conference Papers (Peer reviewed)

[C09] P. Mousavi, L. D. Libera, J. Duret, A. Ploujnikov, *et al.*, "Dasb - discrete audio and speech benchmark," in *Trans. Mach. Learn. Res.*, 2026.

[C08] J. Shi, H.-j. Shim, J. Tian, S. Arora, *et al.*, "VERSA: A versatile evaluation toolkit for speech, audio, and music," in *2025 Annual Conference of the North American Chapter of the Association for Computational Linguistics – System Demonstration Track*, 2025.

[C07] **D. Petermann** and M. Kim, "Hyperbolic distance-based speech separation," in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2024, pp. 1191–1195.

[C06] **D. Petermann**, I. Jang, and M. Kim, "Native multi-band audio coding within hyper-autoencoded reconstruction propagation networks," in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2023, pp. 1–5.

[C05] **D. Petermann**, G. Wichern, A. Subramanian, and J. L. Roux, "Hyperbolic audio source separation," in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2023, pp. 1–5. [Best Student Paper Award & Top 3% Papers]

- [C04] **D. Petermann** and M. Kim, "SpaIn-Net: Spatially-informed stereophonic music source separation," in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2022, pp. 106–110.
- [C03] **D. Petermann**, G. Wichern, Z.-Q. Wang, and J. L. Roux, "The cocktail fork problem: Three-stem audio separation for real-world soundtracks," in *Proc. of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2022, pp. 526–530.
- [C02] **D. Petermann**, S. Beack, and M. Kim, "Harp-net: Hyper-autoencoded reconstruction propagation for scalable neural audio coding," in *Proc. of the IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2021, pp. 316–320.
- [C01] **D. Petermann**, P. Chandna, H. Cuesta, J. Bonada, *et al.*, "Deep learning based source separation applied to choir ensembles," in *Proc. of the International Society for Music Information Retrieval Conference (ISMIR)*, 2020, pp. 733–739.

Journal Articles

- [J03] P. Mousavi†, G. Maimon*, A. Moumen*, **Darius Petermann***, *et al.*, "Discrete audio tokens: More than a survey!, Submitted to Transactions on Machine Learning Research (TMLR)," *Transactions on Machine Learning Research*, 2025, ISSN: 2835-8856.
- [J02] **D. Petermann**, G. Wichern, A. S. Subramanian, Z.-Q. Wang, *et al.*, "Tackling the cocktail fork problem for separation and transcription of real-world soundtracks," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 31, pp. 2592–2605, 2023.
- [J01] P. Chandna, H. Cuesta, **D. Petermann**, and E. Gómez, "A deep-learning based framework for source separation, analysis, and synthesis of choral ensembles," *Frontiers in Signal Processing*, vol. 2, 2022.

Patents

- [P02] S. K. Beack, W. Lim, I. Jang, J. Sung, *et al.*, *Audio signal encoding/decoding methods and apparatus for performing the same*, US Patent App. 63/420 405, 2023.
- [P01] **D. Petermann**, G. Wichern, A. Subramanian, and J. L. Roux, *Audio source separation using hyperbolic embeddings*, US Patent App. 18/191 417, 2023.

4 Honors and Awards

Best Student Paper Award & Top 3% Papers Recognition — Jun 2023
ICASSP 2023
- For the paper entitled "Hyperbolic Audio Source Separation"

Outstanding Reviewer Award Recognition — Jun 2023
ICASSP 2023
- Awarded to the top 5% of the reviewers (220/4445)

"Excellent" Grade with Honor, Music Cognition & Perception — Sep 2020
Universitat Pompeu-Fabra

Magna Cum Laude Honor — May 2016
Berklee College of Music

Dean's List — May 2016 - May 2016
Berklee College of Music
- Appeared on Berklee Dean's List for 7 semesters out of 8

BT Production Award & Scholarship — April 2015
Berklee College of Music
- Award and scholarship from the Electronic Production & Design Dept

5 Tutorials, Other Talks & Posters

Invited talk for the class "Machine Learning for Musicians" Nov 2021
Dept. of Electronic Production & Design (EPD), Berklee College of Music *Boston, MA*

Poster Presentations (Non-Archival) on "Hyperbolic Audio Source Separation" Mar 2023
Luddy AI Center Open House, Indiana University *Bloomington, IN*

6 Certifications

Machine Learning Oct 2019
Stanford - Coursera

Audio Signal Processing for Music Applications Jul 2018
Universitat Pompeu-Fabra - Coursera

7 Technical Skills

Programming Languages: Python, C++, Objective-C, C, Lua, Javascript, LaTeX

Deep Learning (over 8 years of experience): PyTorch, PyTorch Lightning, TensorFlow, JAX

Data: Large-scale multimodal dataset curation and preprocessing (text, audio, vision), experience with Spark, SAIGE, PyTorch DataPipes, distributed training pipelines

Audio Signal Processing (over 12 years of experience): JUCE, MATLAB, Csound, MAX/MSP, PureData

8 Referees

Available upon request.